Fixed-point style index identities: the two formulations are equivalent

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-computability-computability-theory/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: each direction is a complete short construction; nothing is omitted or delegated

Editorial note. The label lem:fixed-equiv names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Universal function and s-m-n theorem. Un(f(x),y) = phi_{f(x)}(y), and s-m-n gives f with phi_{f(x)}(y) = g(x,y) for any partial computable g.. Direction (1)=>(2) unpacks via Un; direction (2)=>(1) manufactures f from g by s-m-n.

Lemma 0.1. *The following statements are equivalent:*

1. *For every partial computable function $g(x, y)$, there is an index e such that for every y,*
$$\varphi_e(y) \simeq g(e, y).$$
2. *For every computable function $f(x)$, there is an index e such that for every y,*
$$\varphi_e(y) \simeq \varphi_{f(e)}(y).$$

Proof. (1) $\Rightarrow$ (2): Given f, define g by $g(x, y) \simeq \mathrm{Un}(f(x), y)$. Use (1) to get an index e such that for every y,

$$\begin{aligned}\varphi_e(y) &= \mathrm{Un}(f(e), y)\\ &= \varphi_{f(e)}(y).\end{aligned}$$

(2) $\Rightarrow$ (1): Given g, use the s-m-n theorem to get f such that for every x and y, $\varphi_{f(x)}(y) \simeq g(x, y)$. Use (2) to get an index e such that

$$\begin{aligned} \varphi_e(y) &= \varphi_{f(e)}(y) \\ &= g(e, y). \end{aligned}$$

This concludes the proof. $\square$